# LUKAS FLEURY

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## SKILLS

- **CAD & Design:** SolidWorks (CSWP), FEA, AutoCAD, GD&T (ASME Y14.5), DFM, Mastercam
- **Software & Programming:** MoveIt, Python, C++, RViz, PDM, SAP, MATLAB

## WORK EXPERIENCE

**LiftWerx | *Cambridge, Ontario* | Operations Engineering Co-op** *08/2026 - Present*

- Responsible for the **execution** of **turbine repair operations** and lift plans for 12 field crews, applying engineering principles to **balance safety, efficiency, logistics, and sustainability** across active wind farms.
- **Designed** a **yaw drive extractor device** to remove failed turbine yaw drives, using **hydraulics jacks** braced on the nacelle bed plate to overcome M20 bolt removal in a confined space.
- Contribute to the **design** and **implementation** of accessory **crane lifting equipment** and rigging systems, enabling major turbine components to be serviced without full crane teardown.

**Mammoet | *Puslinch, Ontario* | Engineering Co-op** *01/2026 - 05/2026*

- **Produced and revised 80+** fabrication drawings for **crane and heavy-lift equipment**, improving drawing accuracy, standardization, and **database integration**.
- **Designed** detailed usage, assembly, and setup drawings for heavy-lift transport systems (jack-and-slide, SPMT interfaces), reducing field complexity and supporting safer, faster on-site execution.
- **Developed** project drawings to install a **150-ton** transformer at Bruce Power, and **100-ton** steam generators at Pickering Nuclear.

## TECHNICAL PROJECTS

**8” Autonomous Waypoint Drone | Design, PID Tuning + Autonomous Navigation**

- **Designed** and **built** a custom FPV-style drone, implementing the **mechanical frame, electrical systems**, soldering connections, and **custom tuned** a **flight controller** within a fixed cost constraint.
- **Integrated** the full stack (SpeedyBee F405 AIO, RS2205 motors, FlySky, 3S LiPo, GPS, Radio, FPV Camera), then PID-tuned in Betaflight across crash-fix-fly cycles to reach **stabilized flight.**
- Implemented **GPS radio telemetry** and ArduPilot to enable autonomous **waypoint navigation**.

**5-DOF 20” Radius Robotic Arm | Design, Actuation + Motion Planning (In Progress)**

- **Actuated** by **NEMA 17** stepper **motors** driven through CNC shield controlled drivers, with a **custom 28:1 planetary gear reduction box** built for each joint to **boost torque and holding capacity.**
- Conducted **torque load tests** on the geared joints to **validate actuator sizing** and confirm the arm could hold and move payloads without stalling or back-driving.
- *(In progress)* Implementing **ROS 2 (MoveIt, RViz)** for visual pose control and collision-free **path planning.**

**OYAP Student Manufacturing Projects**

- **Developed** and **incorporated** challenging multi-part manufacturing projects to the **OYAP** curriculum.
- **Created** advanced projects that improved technical skills and gave **80+ students** experience in interpreting advanced blueprints while reinforcing **precise machining operations** on the lathe, mill, and CNC machines.

## AWARDS

- 1st Place – Waterloo Regional Precision Machining Competition | SkillsOntario
- Engineering Certificate Program (ECP) | St. Benedicts C.S.S.

## EDUCATION

*Expected April 2030*

**University of Waterloo**

Bachelor of Applied Science, Mechanical Engineering